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**SOFTWARE REQUIREMENTS SPECIFICATION (SRS)  
ON  
TransGraph-PPI: A Hybrid Ensemble Framework for  
Sequence-Based Protein–Protein Interaction Prediction**

*Submitted in partial fulfillment of the requirements for the degree*

**BACHELOR OF ENGINEERING  
in  
COMPUTER SCIENCE & ENGINEERING**

*Submitted by*

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**2026-27**

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## Chapter 1

# Software Requirements Specification (SRS)

## 1.1 Introduction

### 1.1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for TransGraph-PPI: A Hybrid Ensemble Framework for Sequence-Based Protein–Protein Interaction Prediction. This document is intended for the development team, project guide, and evaluators. It serves as the primary reference for system design, development, testing, and evaluation.

### 1.1.2 Scope

TransGraph-PPI is a computational bioinformatics software system designed to predict protein–protein interactions (PPIs) from amino acid sequences using a hybrid deep learning framework. The system integrates pre-trained ESM-2 protein language model embeddings with a Graph Attention Network (GAT) and combines their outputs using a stacking-based ensemble (XGBoost meta-learner). The system further provides SHAP-based model interpretability and network centrality analysis for drug target identification. A web-based interface allows researchers to submit protein sequence pairs and visualize predicted interaction networks.

### 1.1.3 Definitions, Acronyms, and Abbreviations

| Term / Acronym | Definition                                                                  |
|----------------|-----------------------------------------------------------------------------|
| PPI            | Protein–Protein Interaction                                                 |
| ESM-2          | Evolutionary Scale Modeling v2 – a transformer-based protein language model |
| GAT            | Graph Attention Network – a GNN using attention mechanisms                  |
| GNN            | Graph Neural Network                                                        |
| PLM            | Protein Language Model                                                      |
| SHAP           | SHapley Additive exPlanations – an interpretability framework               |
| XGBoost        | Extreme Gradient Boosting – ensemble machine learning algorithm             |
| ROC-AUC        | Area Under the Receiver Operating Characteristic Curve                      |
| PR-AUC         | Area Under the Precision-Recall Curve                                       |
| STRING         | Search Tool for the Retrieval of Interacting Genes/Proteins database        |
| BioGRID        | Biological General Repository for Interaction Datasets                      |
| UniProt        | Universal Protein Resource database                                         |
| ChEMBL         | Chemical database of bioactive molecules with drug-like properties          |

| Term /<br>Acronym | Definition                          |
|-------------------|-------------------------------------|
| SRS               | Software Requirements Specification |
| API               | Application Programming Interface   |
| MLP               | Multi-Layer Perceptron              |

#### 1.1.4 Overview

This SRS is organized as follows: Section 3.1 provides an introduction. Section 3.2 gives an overall system description. Section 3.3 specifies functional requirements. Section 3.4 specifies non-functional requirements. Section 3.5 describes the system architecture. Section 3.6 describes the user interface design. Section 3.7 lists hardware and software requirements. Section 3.8 defines performance requirements. Section 3.9 specifies design constraints. Section 3.10 covers other requirements.

## 1.2 Overall Description

### 1.2.1 Product Perspective

TransGraph-PPI is a standalone bioinformatics software system with a web-based front-end and a Python-based computational back-end. It interfaces with public protein databases (STRING, BioGRID, UniProt, ChEMBL) for data collection and uses HuggingFace-hosted ESM-2 model weights for embedding generation. The system is designed as a research tool for bioinformaticians, computational biologists, and drug discovery researchers.

### 1.2.2 Product Functions

The main functions of the system are:

- Data collection and preprocessing of human protein interaction pairs from public databases.
- Sequence embedding generation using the pre-trained ESM-2 protein language model.
- Protein interaction graph construction with ESM-2 node features.
- Base learner 1: ESM-2 + MLP classifier for sequence-based PPI prediction.
- Base learner 2: GAT-based link prediction on the protein interaction graph.
- Stacking ensemble: XGBoost meta-learner combining outputs of both base learners.
- SHAP-based interpretability analysis of model predictions.
- Network analysis and drug target identification using centrality measures.
- Web-based interface for interactive PPI prediction and visualization.

### 1.2.3 User Classes and Characteristics

| User Class                    | Technical Level | Primary Interaction                                                                                     |
|-------------------------------|-----------------|---------------------------------------------------------------------------------------------------------|
| Bioinformatician / Researcher | High            | Full system usage: data pipeline, model training, interpretability analysis, drug target identification |
| Drug Discovery Scientist      | Medium          | Web interface for PPI prediction, interaction network visualization, and drug target reports            |
| CS Student / Developer        | High            | Model implementation, training, evaluation, and integration of new components                           |
| System Administrator          | High            | Deployment, maintenance, and system configuration                                                       |

#### **1.2.4 Assumptions and Dependencies**

- Protein interaction data from STRING and BioGRID is assumed to be experimentally validated and available for download.
- ESM-2 pre-trained model weights are accessible through HuggingFace Transformers.
- Internet connectivity is required for database access during the data collection phase.
- NVIDIA GPU availability is assumed for efficient model training (CPU fallback supported but significantly slower).
- Python 3.9+ and required libraries must be installed.

### 1.3 Functional Requirements

**FR-01: Data Collection Module.** The system shall collect experimentally validated human PPI pairs from STRING and BioGRID databases. The system shall retrieve corresponding protein sequences from UniProt. The system shall collect drug target information from ChEMBL for target prioritization validation. Both positive (interacting) and negative (non-interacting) protein pairs shall be included.

**FR-02: Data Preprocessing Module.** The system shall clean protein sequences by removing incomplete, duplicate, or redundant entries. The system shall handle class imbalance using weighted loss functions and/or appropriate sampling techniques. The system shall split data into training, validation, and test sets with no data leakage between sets.

**FR-03: Feature Extraction Module (ESM-2 Embeddings).** The system shall use the pre-trained ESM-2 protein language model to generate fixed-length dense embedding representations for each protein sequence. Embeddings shall capture contextual, evolutionary, and biochemical properties from raw sequence data. For each protein pair, embeddings shall be concatenated to form the input representation.

**FR-04: Base Learner 1 – ESM-2 + MLP.** The system shall implement an MLP classifier that takes concatenated ESM-2 embeddings of a protein pair as input and produces an interaction probability as output. The MLP shall include appropriate hidden layers, dropout regularization, and a sigmoid output activation. The model shall be trained using binary cross-entropy loss with class weighting.

**FR-05: Graph Construction Module.** The system shall construct a protein interaction graph where proteins are represented as nodes (initialized with ESM-2 embeddings) and known interactions form edges. The graph shall be constructed from verified PPI data and stored in a format compatible with PyTorch Geometric.

**FR-06: Base Learner 2 – Graph Attention Network.** The system shall implement a Graph Attention Network (GAT) applied to the protein interaction graph. The GAT shall use multi-head attention to learn importance-weighted aggregation of neighbouring node features. Learned node embeddings shall be combined (e.g., element-wise product and concatenation) to compute interaction probabilities for link prediction.

**FR-07: Ensemble Module.** The system shall implement a soft voting ensemble combining predicted probabilities from both base models. The system shall additionally implement a stacking-based meta-learner using XGBoost trained on the probability outputs of both base models. The ensemble shall be trained using cross-validation to avoid data leakage.



**FR-08: Interpretability Module.** The system shall apply SHAP (SHapley Additive exPlanations) analysis to interpret the contributions of individual models and features to final predictions. The system shall generate SHAP summary plots, feature importance rankings, and attention-weight visualizations from the GAT component.

**FR-09: Performance Evaluation Module.** The system shall evaluate model performance using: Accuracy, Precision, Recall, F1-score, ROC-AUC, and PR-AUC. Cross-validation techniques shall be applied to ensure robustness and prevent data leakage. The ensemble framework shall be compared against individual base models and traditional ML baselines (Random Forest with handcrafted features).

**FR-10: Network Analysis and Drug Target Identification Module.** The system shall analyze the predicted protein interaction network using centrality-based measures (degree centrality, betweenness centrality, closeness centrality, PageRank). The system shall cross-reference identified hub proteins with ChEMBL and DrugBank to identify and prioritize potential therapeutic target proteins. The system shall generate a ranked list of drug target candidates with supporting evidence.

**FR-11: Web Interface.** The system shall provide a web-based platform for interactive PPI prediction. Users shall be able to submit protein sequence pairs (in FASTA format) and receive interaction probability predictions. The system shall visualize the predicted interaction network as an interactive graph. Results shall be exportable in CSV and JSON formats.

## 1.4 Non-Functional Requirements

**NFR-01: Performance.** The system shall process a single protein pair query through the trained ensemble model and return a prediction within 10 seconds on the recommended hardware configuration. Batch prediction for up to 1,000 protein pairs shall complete within 30 minutes on GPU hardware.

**NFR-02: Scalability.** The system architecture shall support processing of protein interaction datasets containing up to 500,000 protein pairs. The data pipeline shall be modular, allowing addition of new database sources without significant architectural changes.

**NFR-03: Reliability.** The system shall produce consistent, deterministic results for the same input given fixed random seeds. Model checkpointing shall be implemented to allow training recovery from interruptions. The system shall handle malformed or incomplete protein sequences gracefully with appropriate error messages.

**NFR-04: Maintainability.** The codebase shall be modular, with separate components for data processing, model training, evaluation, interpretability, and web interface. Code shall be documented with docstrings and inline comments. Unit tests shall cover critical data processing and model evaluation functions.

**NFR-05: Usability.** The web interface shall be intuitive and require no programming knowledge for basic prediction tasks. API documentation shall be provided for programmatic access. Results shall be presented with clear visualizations and interpretable explanations.

**NFR-06: Security.** The web interface shall validate all user inputs to prevent injection attacks. Uploaded protein sequences shall not be stored permanently beyond the session. API access shall be rate-limited to prevent abuse.

## 1.5 System Architecture

This section presents the overall system architecture of TransGraph-PPI. The architecture diagram (Figure 1.1) illustrates the complete end-to-end data flow—from raw protein sequence input through dual-branch feature extraction, stacking ensemble prediction, post-prediction biological analysis, and interactive web-based visualization.

### 1.5.1 Architecture Diagram

### 1.5.2 Tier-wise Architecture Description

The TransGraph-PPI architecture is structured into five logically distinct processing tiers.

**Tier 1 — Input Layer:** The system accepts a protein pair (Protein A and Protein B) in FASTA format as the shared input to two independent parallel processing branches, enabling simultaneous extraction of sequence-level semantics and network-level topology.

**Tier 2 — Dual-Branch Feature Extraction:** The left branch passes each sequence through the ESM-2 Transformer encoder producing dense fixed-length embeddings; these are concatenated and fed to an MLP classifier. The right branch constructs a Protein Interaction Graph from STRING/BioGRID and applies a GAT for link prediction via attention-weighted aggregation of neighbour node features.

**Tier 3 — XGBoost Stacking Ensemble:** The MLP probability and GAT link prediction score are concatenated and passed to an XGBoost meta-learner that learns a non-linear, data-driven combination, producing the Final Interaction Probability Score.

**Tier 4 — Post-Prediction Analysis:** SHAP Explainability provides residue-level feature attribution. Mutation Impact Scanning evaluates amino acid substitutions. Subcellular Localization Filtering removes biologically implausible predictions. Pathway Analysis maps interactions to KEGG/Reactome. Network Centrality Detection identifies hub proteins. Drug Target Mapping cross-references hubs with ChEMBL.

**Tier 5 — Interactive Web Platform:** Outputs are presented via an Interactive PPI Network Visualization, 3D AlphaFold Structure Viewer, Dynamic Interaction Graph explorer, and a comprehensive Result Dashboard.

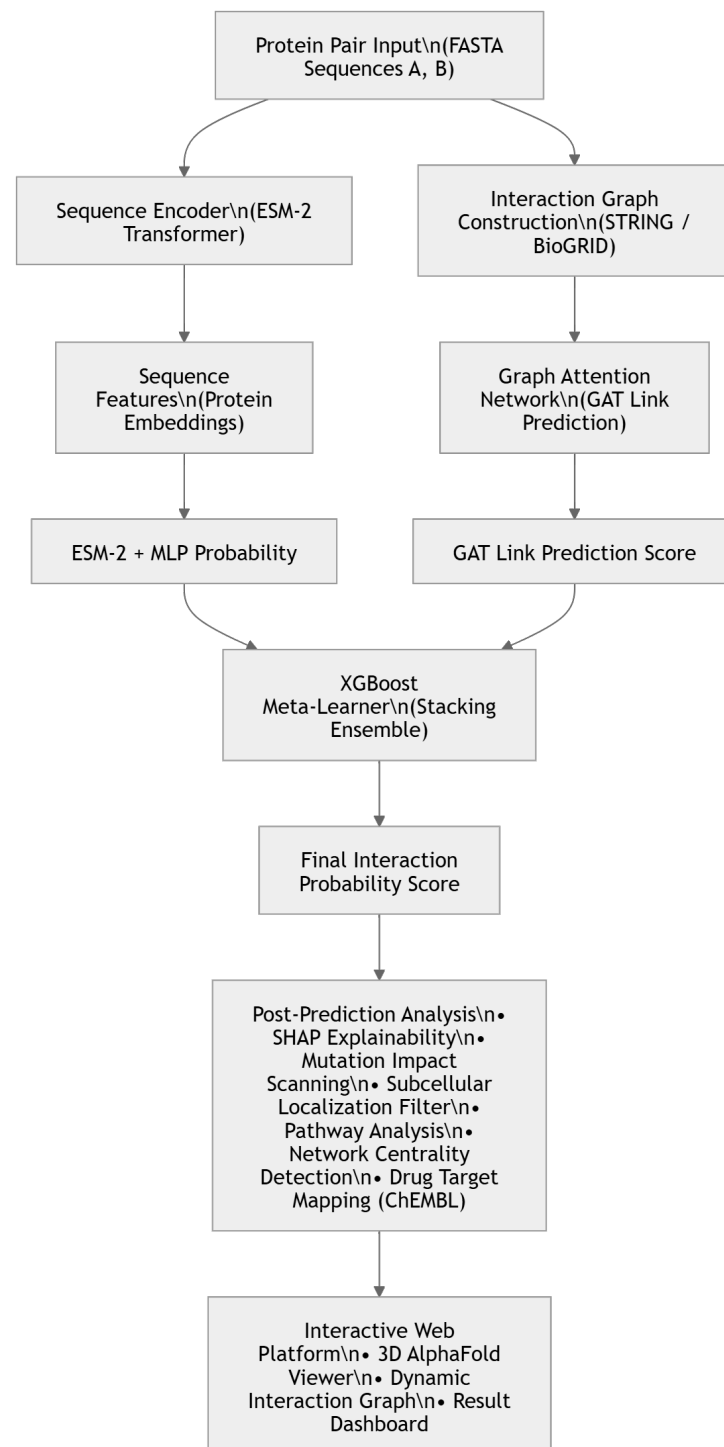


Figure 1.1: System Architecture of TransGraph-PPI — End-to-End Prediction and Analysis Pipeline

## **1.6 User Interface Design**

### **1.6.1 Input Interface**

The web interface shall provide: (1) a text input field accepting protein sequences in raw or FASTA format; (2) an option to submit a single pair or batch CSV upload; (3) a model selection dropdown (ESM-2 + MLP, GAT, Ensemble); (4) a submit button triggering prediction.

### **1.6.2 Results Display Interface**

The results page shall display: (1) the predicted interaction probability (0–1 scale) with a confidence indicator; (2) a binary interaction prediction (Interacting / Non-Interacting) based on a configurable threshold (default: 0.5); (3) a SHAP feature importance chart; (4) an interactive protein interaction network graph using NetworkX / D3.js visualization.

### **1.6.3 Network Analysis Interface**

A separate network analysis dashboard shall display: (1) the full predicted PPI network with node colouring by centrality score; (2) a ranked table of potential drug target proteins with centrality scores and drug database cross-references; (3) export buttons for network data in GraphML and CSV formats.

## 1.7 Hardware and Software Requirements

### 1.7.1 Hardware Requirements

| Component         | Specification                                                                      |
|-------------------|------------------------------------------------------------------------------------|
| Processor         | Intel Core i5/i7 (8th Gen or higher) or AMD Ryzen 5/7                              |
| RAM               | Minimum 8 GB (16 GB recommended for model training)                                |
| Storage           | 20–30 GB free space (SSD recommended for faster data I/O)                          |
| GPU (Recommended) | NVIDIA GPU with CUDA support (e.g., GTX 1060 / RTX 3060 or higher, min. 6 GB VRAM) |
| Operating System  | Windows 10/11 (64-bit) or Linux Ubuntu 20.04/22.04                                 |

### 1.7.2 Software Requirements

| Category               | Tool / Library                        | Purpose                            |
|------------------------|---------------------------------------|------------------------------------|
| Programming Language   | Python 3.9+                           | Core development language          |
| Development Env.       | Jupyter Notebook, PyCharm, VS Code    | IDE and interactive development    |
| Deep Learning          | PyTorch 2.0+                          | Model training and inference       |
| Protein Language Model | HuggingFace Transformers              | ESM-2 model loading and inference  |
| Graph Neural Networks  | PyTorch Geometric                     | GAT implementation                 |
| Ensemble Learning      | XGBoost, Scikit-learn                 | Meta-learner and baseline models   |
| Explainability         | SHAP, Captum                          | Model interpretability             |
| Data Processing        | NumPy, Pandas                         | Data manipulation                  |
| Visualization          | Matplotlib, Seaborn, NetworkX, Plotly | Plotting and network visualization |
| Web Framework          | Flask / FastAPI                       | Web interface backend              |
| Database Sources       | STRING, BioGRID, UniProt, ChEMBL      | PPI data and drug targets          |
| Version Control        | Git / GitHub                          | Source code management             |

## 1.8 Performance Requirements

| Metric                              | Target Value         | Measurement Method      |
|-------------------------------------|----------------------|-------------------------|
| PPI Prediction Accuracy (Ensemble)  | $\geq 0.90$          | 5-fold cross-validation |
| ROC-AUC (Ensemble)                  | $\geq 0.95$          | 5-fold cross-validation |
| PR-AUC (Ensemble)                   | $\geq 0.92$          | 5-fold cross-validation |
| F1-Score (Ensemble)                 | $\geq 0.88$          | 5-fold cross-validation |
| Improvement over single models      | $\geq 2\%$ F1 gain   | Comparison table        |
| Single pair prediction time (web)   | $\leq 10$ seconds    | End-to-end latency test |
| Batch prediction (1,000 pairs, GPU) | $\leq 30$ minutes    | Timed batch experiment  |
| Model training time (GPU)           | $\leq 4$ hours/model | GPU training log        |

## **1.9 Design Constraints**

### **1.9.1 Sequence-Only Prediction**

The system must operate using only protein amino acid sequences as input, without requiring experimentally determined or computationally predicted 3D structures. This ensures broad applicability across the entire proteome where structural data is unavailable.

### **1.9.2 Pre-trained Model Dependency**

The system shall use ESM-2 as the primary protein language model. The model weights shall be loaded from HuggingFace and used in inference mode (frozen) during the embedding extraction phase, with only the downstream MLP head trained from scratch.

### **1.9.3 Framework Compatibility**

All model components shall be implemented in PyTorch. Graph neural network components shall use PyTorch Geometric. All code shall be compatible with Python 3.9+ on both Windows and Linux platforms.

### **1.9.4 Data Privacy and Ethics**

The system shall use only publicly available, ethically sourced protein interaction datasets. No private or proprietary biological data shall be used. The web interface shall not permanently store submitted protein sequences.

### **1.9.5 Reproducibility**

All experiments shall use fixed random seeds for reproducibility. Dataset splits shall be saved and documented. Model hyperparameters and training configurations shall be stored in configuration files (YAML/JSON).



## **1.10 Other Requirements**

### **1.10.1 Documentation Requirements**

The system shall include: (1) a comprehensive README with installation instructions; (2) API documentation for all public functions; (3) a user manual for the web interface; (4) a technical report detailing model architecture, training procedure, and evaluation results.

### **1.10.2 Testing Requirements**

The system shall undergo: (1) unit testing for all data processing and evaluation functions; (2) integration testing for the end-to-end prediction pipeline; (3) performance benchmarking against baseline models on standard datasets; (4) user acceptance testing of the web interface.

### **1.10.3 Deployment Requirements**

The web interface shall be deployable on a standard Linux server with Python 3.9+ and the required dependencies. A Docker container configuration shall be provided for consistent deployment. The application shall be accessible via a standard web browser without additional client-side software installation.

### **1.10.4 Long-Term Maintainability**

The modular architecture shall allow future extension with new protein language models, additional graph neural network variants, or updated drug target databases without requiring complete system redesign. Regular updates to database connections shall be planned to maintain access to the latest PPI data.